

MUCH ADO ABOUT NULL THINGS

A replication issue in spatial stats

JEFF SAUER
SERGIO REY

TAYLOR OSHAN
LEVI JOHN WOLF



LOCAL STATISTICS

A REPLICATION FAILURE

“NULLS” IN LOCAL TESTS

The importance of null hypotheses in local statistics

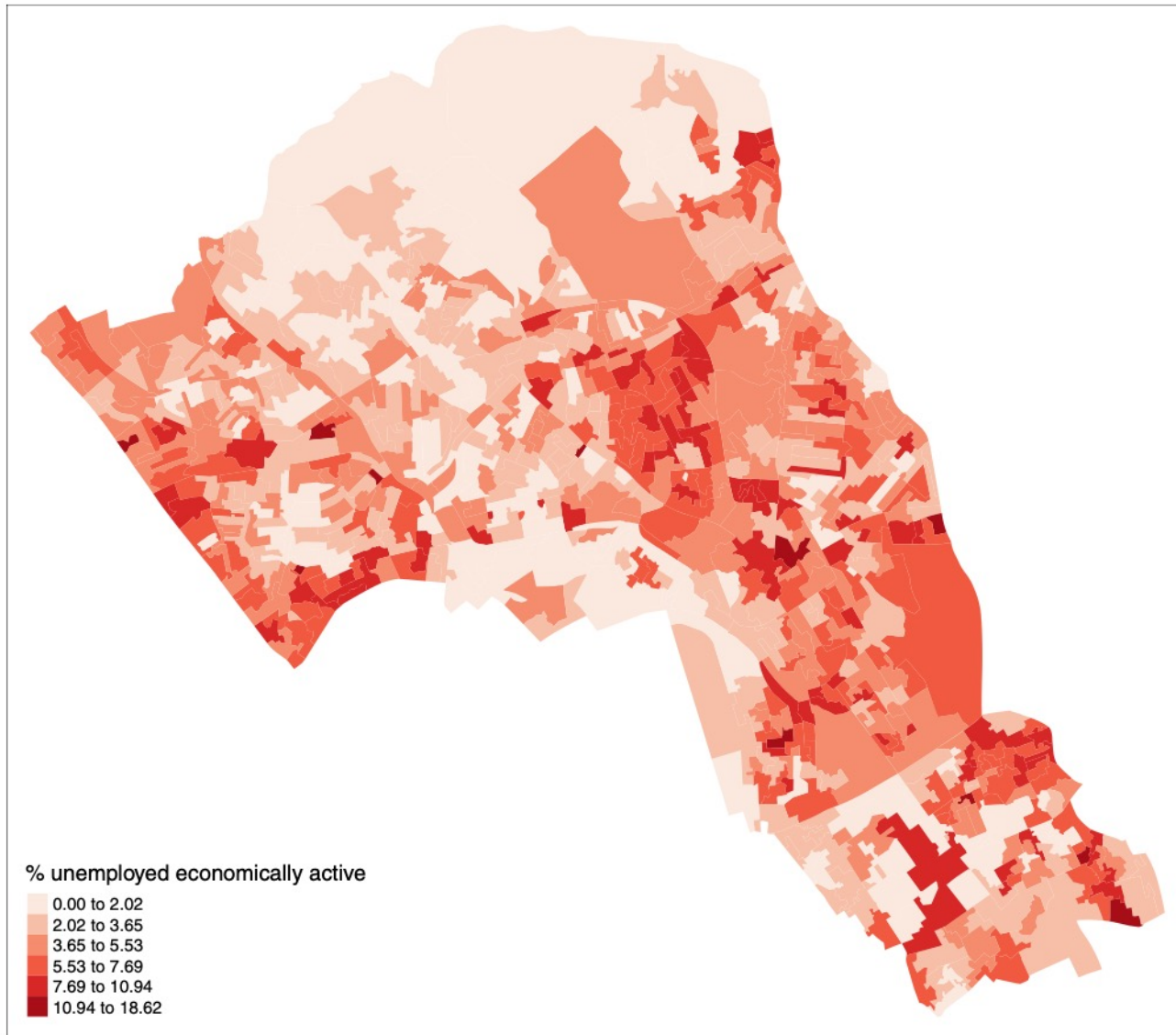
LOCAL STATISTICS

FORMALIZING INTUITION

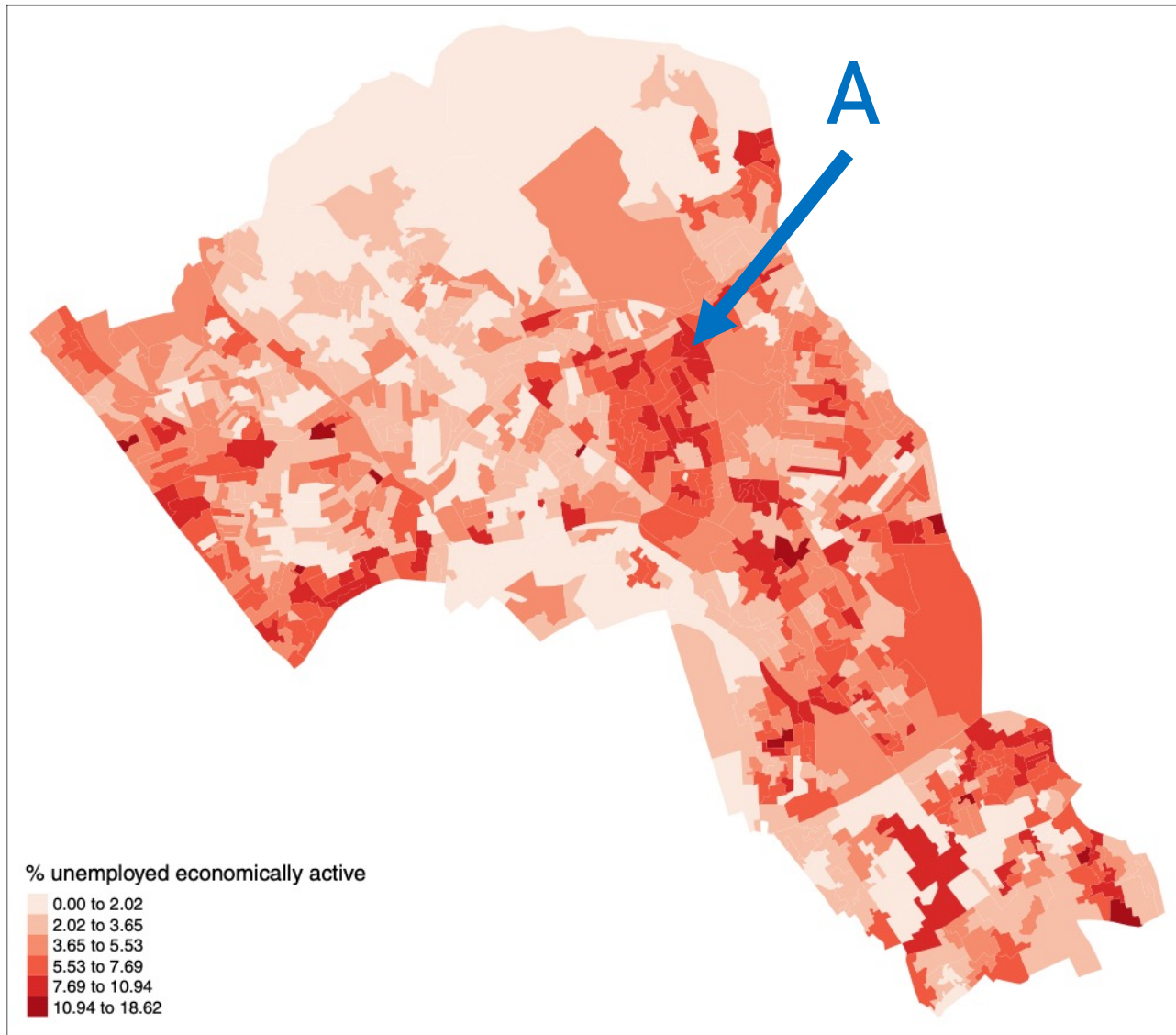
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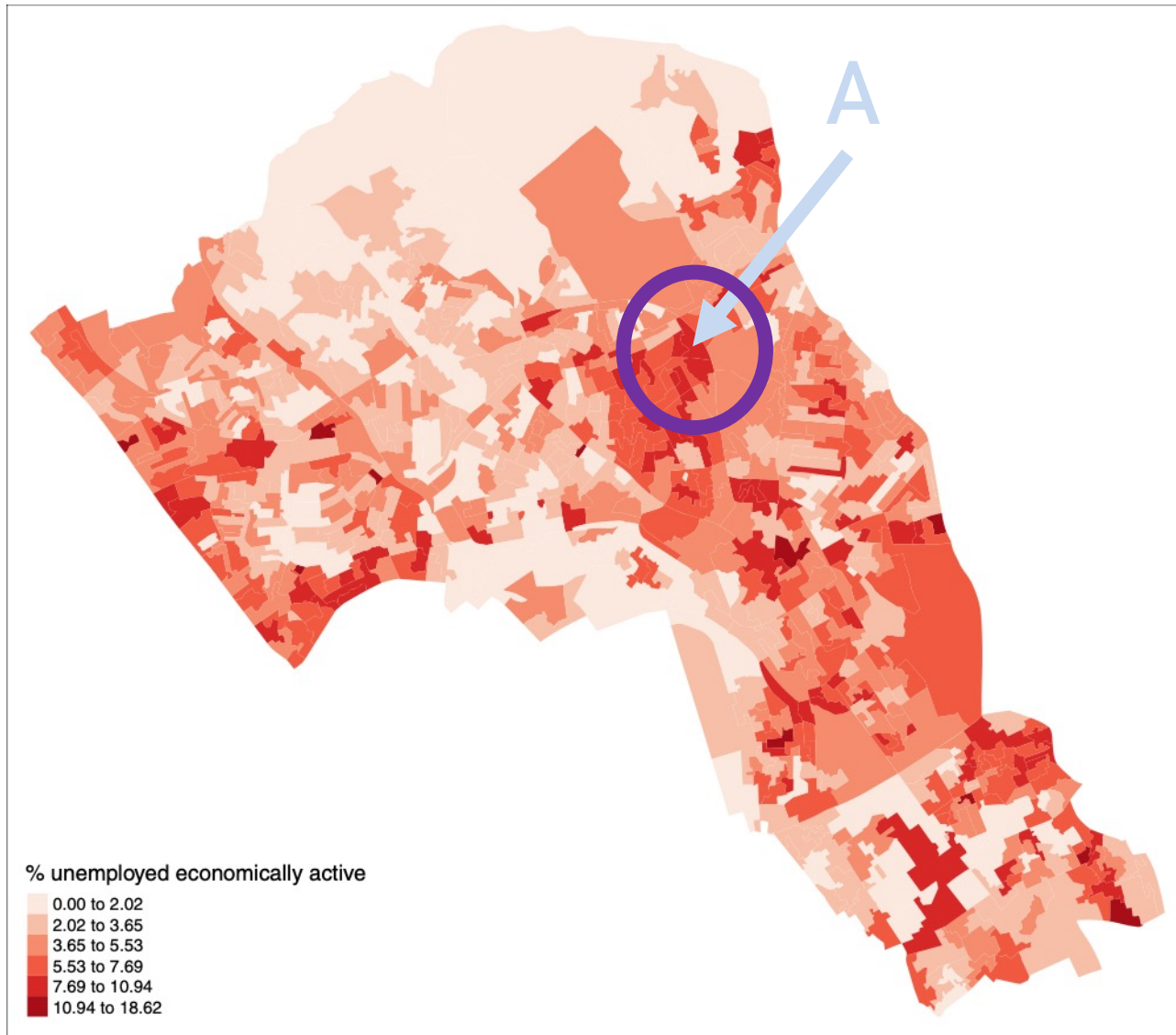
The importance of null hypotheses in local statistics



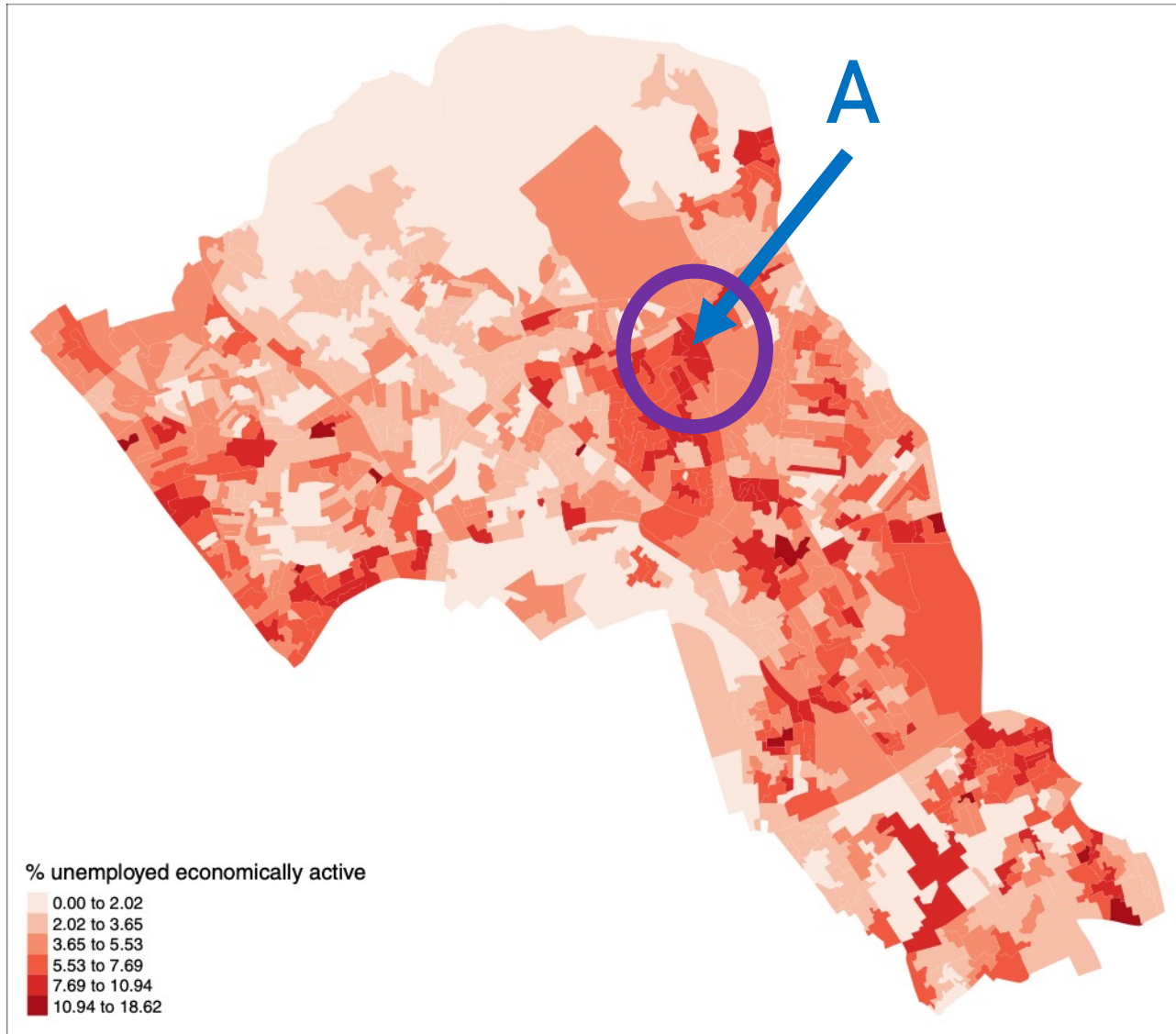
Local statistics: find outliers and/or clusters



Local statistics: find outliers and/or clusters

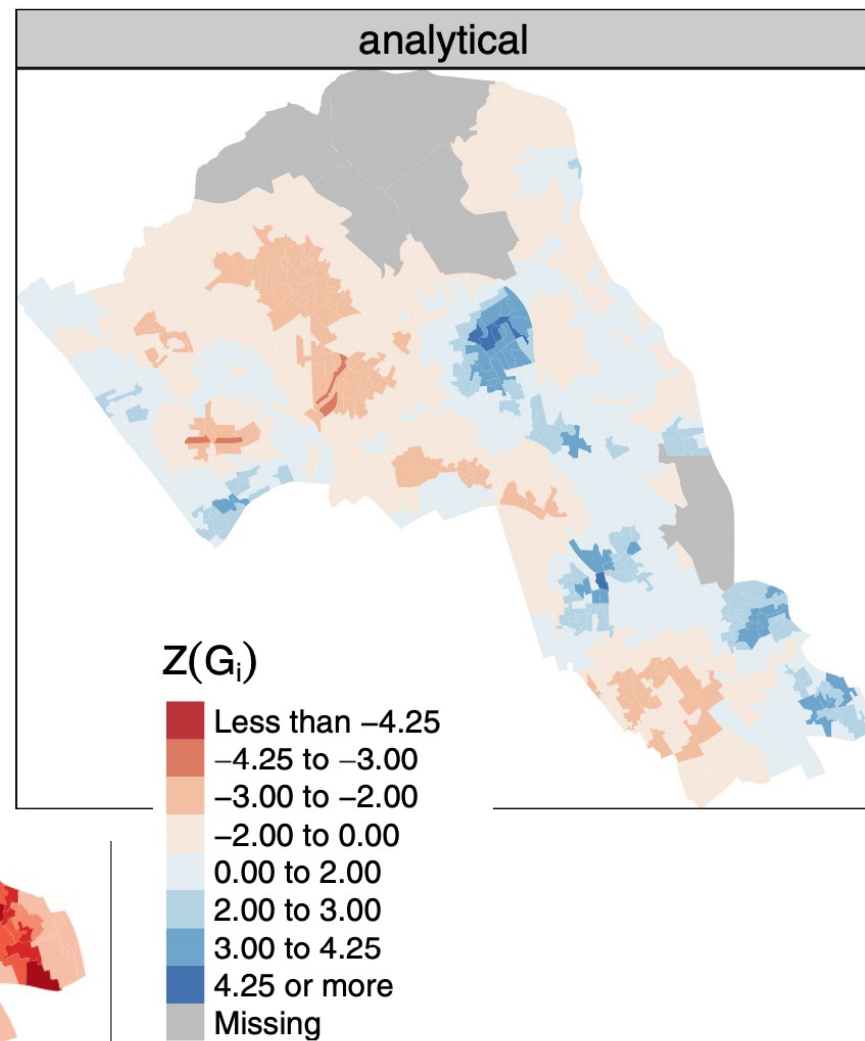
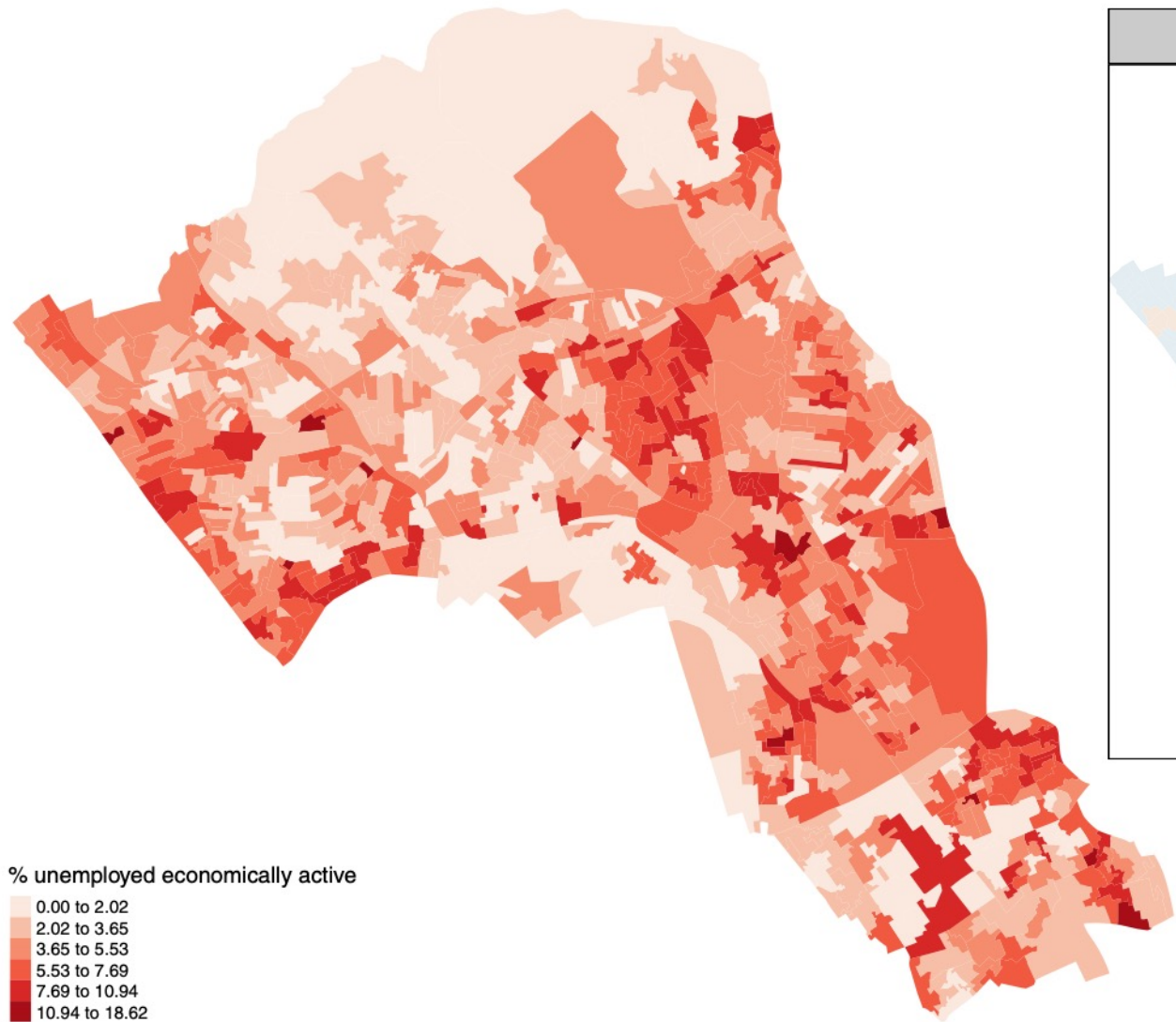


Local statistics: find outliers and/or clusters

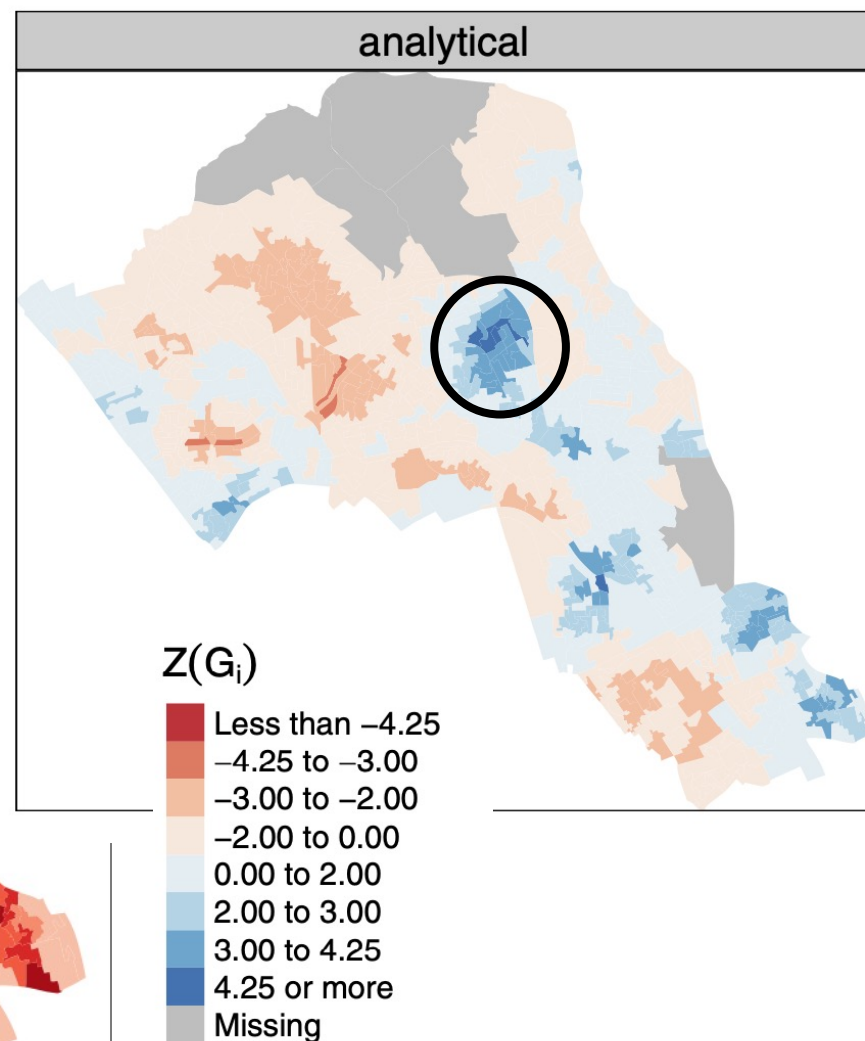
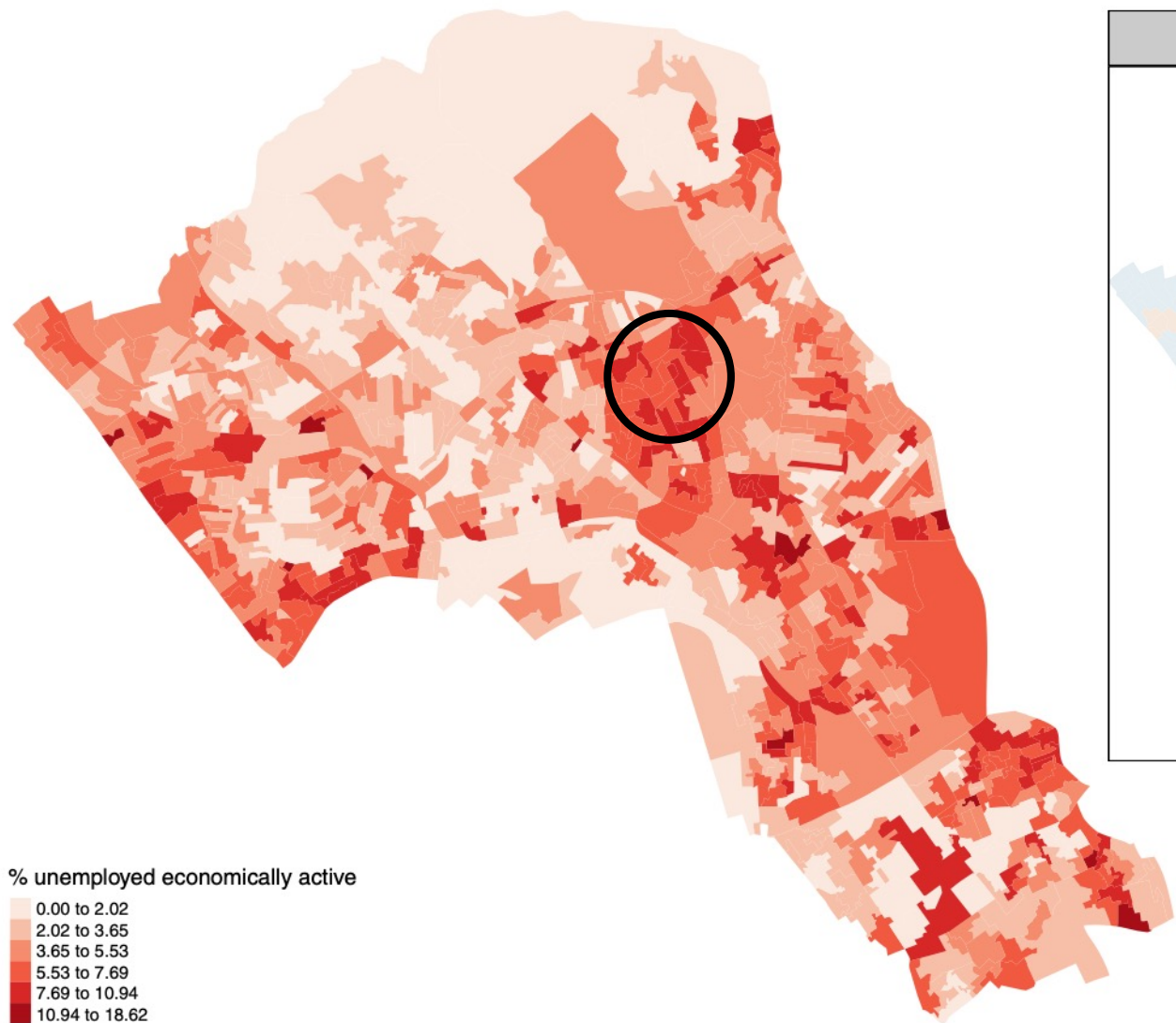


Is the
distribution
around this site
what we'd
expect?

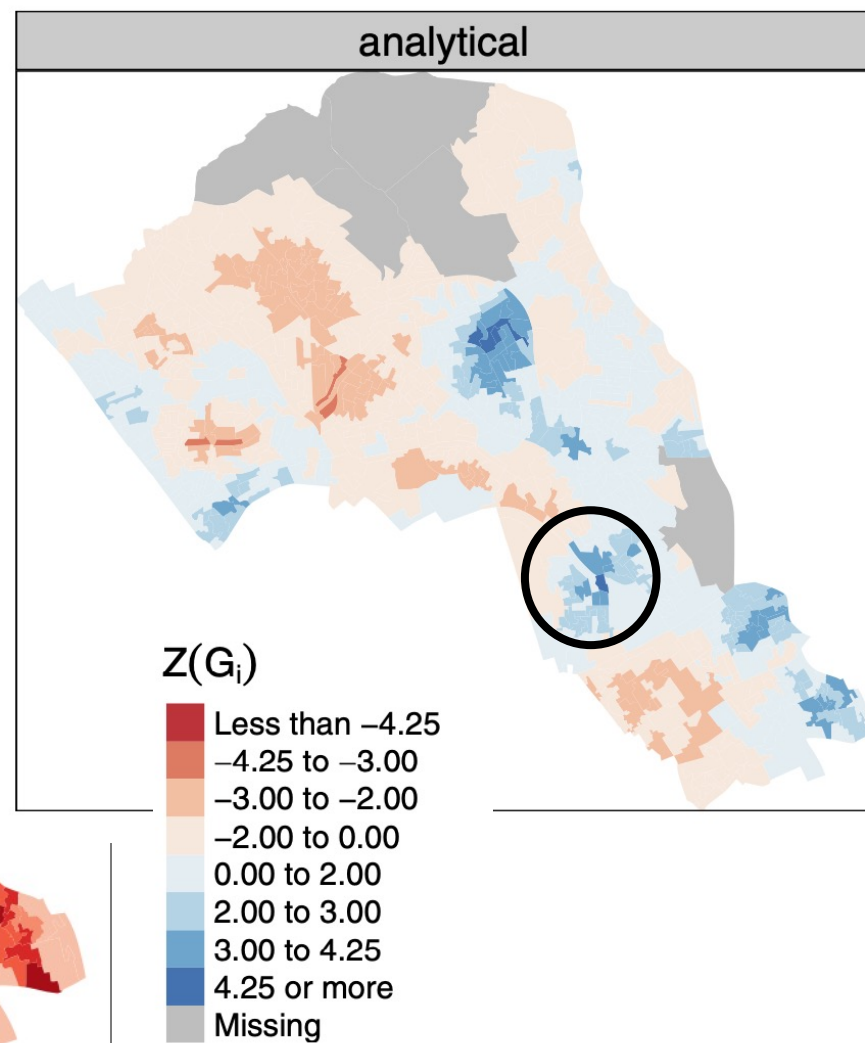
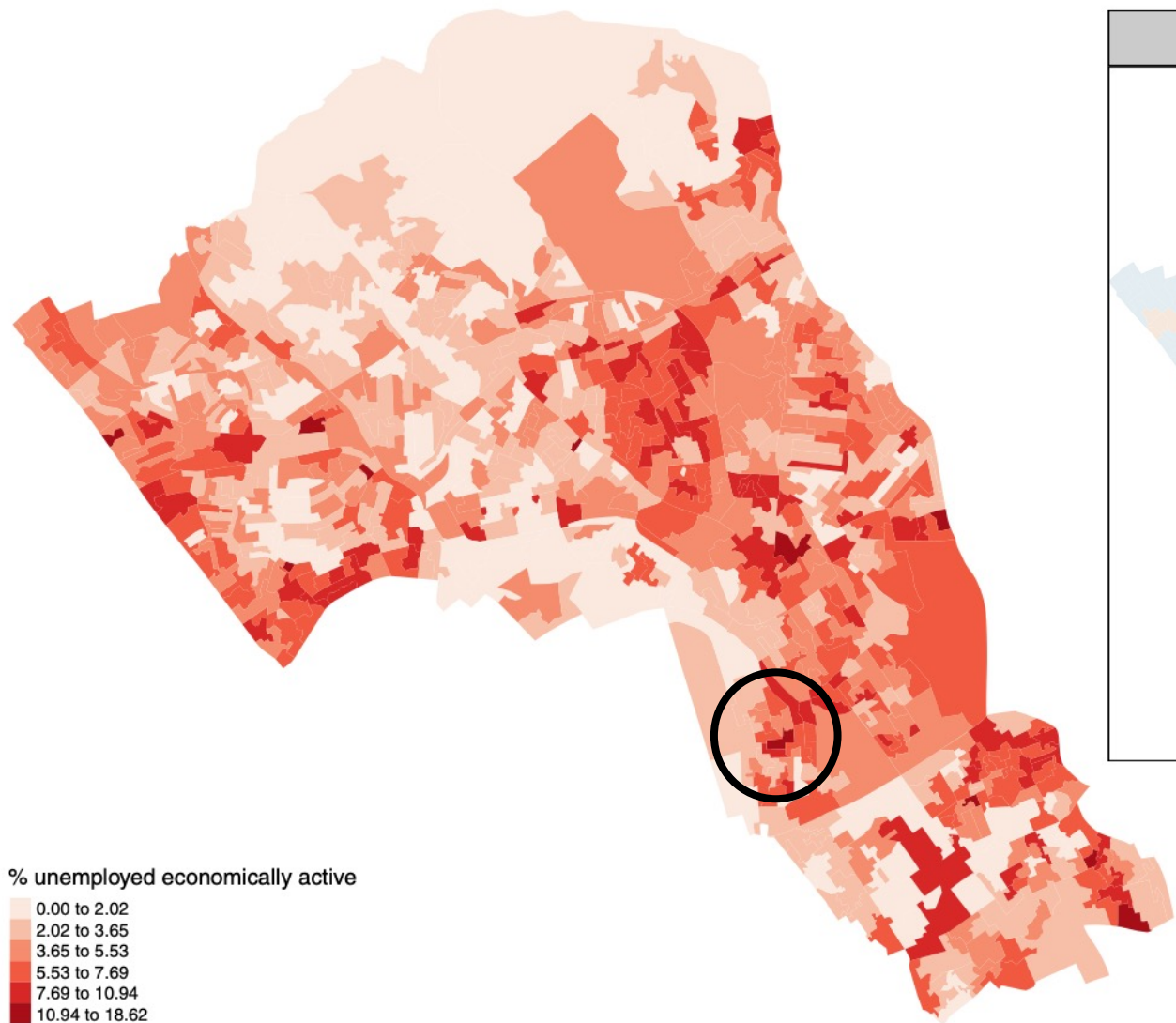
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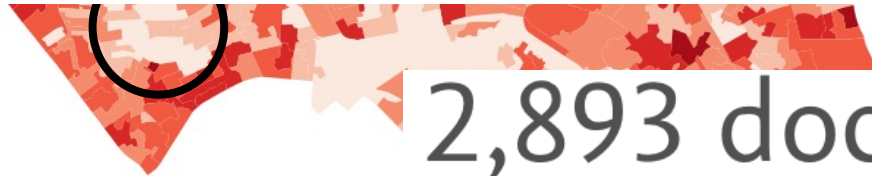
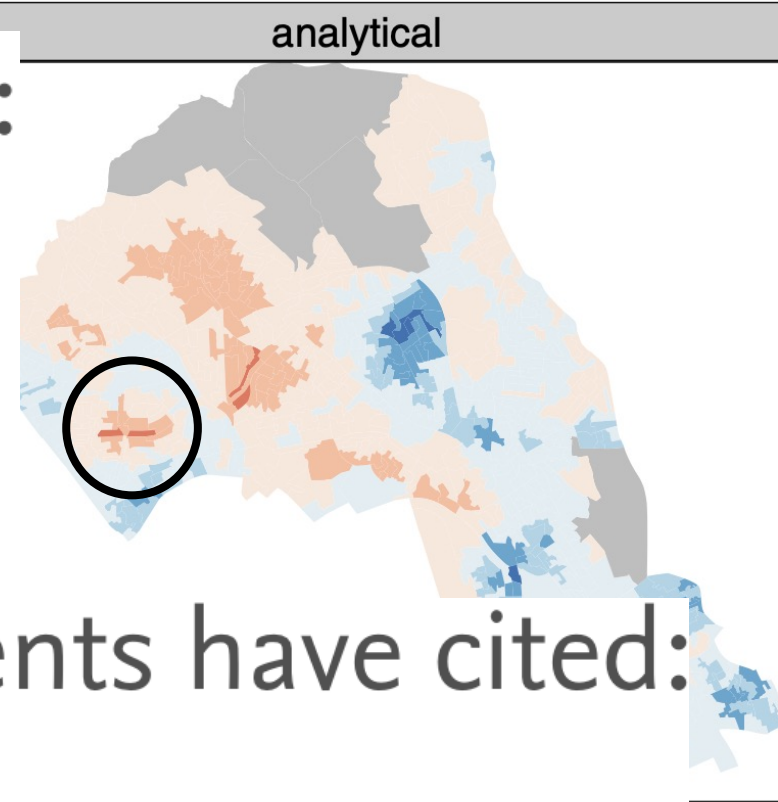
Local statistics: find outliers and/or clusters

5,679 documents have cited:

Local Indicators of Spatial Association—LISA

Anselin L.

(1995) Geographical Analysis, 27 (2) , pp. 93-115.



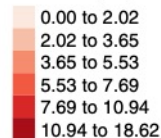
2,893 documents have cited:

The Analysis of Spatial Association by Use of Distance Statistics

Getis A., Ord J.K.

(1992) Geographical Analysis, 24 (3) , pp. 189-206.

% unemployed economically active



Local statistics: find outliers and/or clusters

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TEST (2018) 27:716–748

<https://doi.org/10.1007/s11749-018-0599-x>

ORIGINAL PAPER



CrossMark

Comparing implementations of global and local indicators of spatial association

Roger S. Bivand¹  · David W. S. Wong²

Received: 4 April 2018 / Accepted: 14 July 2018 / Published online: 27 July 2018

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doi.org/10.1007/s11749-018-0599-x

Comparing implementations of global and local indicators of spatial association

Roger S. Bivand¹  · David W. S. Wong²

User choices for local measures both of software and of inferential method over and above the handling of multiple comparisons will have consequences for conclusions

doi.org/10.1007/s11749-018-0599-x

Comparing implementations of global and local indicators of spatial association

Roger S. Bivand¹  · David W. S. Wong²

[T]he spatial patterns of Z values generated by conditional permutation for local measures differ considerably from those calculated using analytical methods.

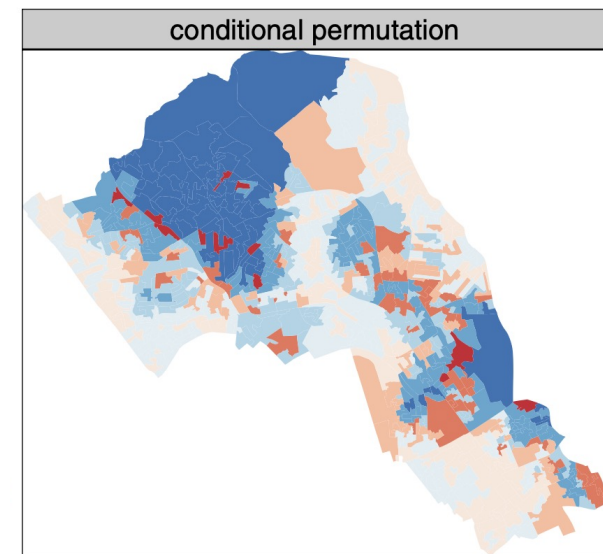
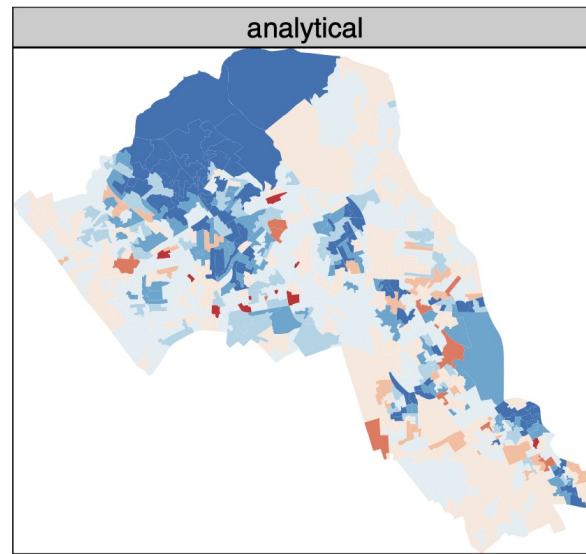
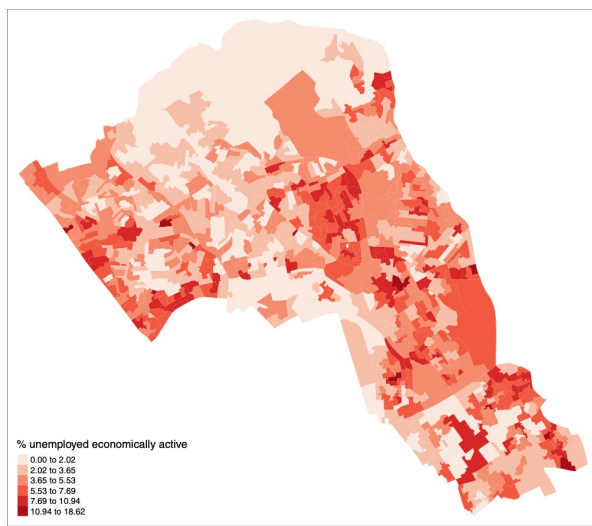
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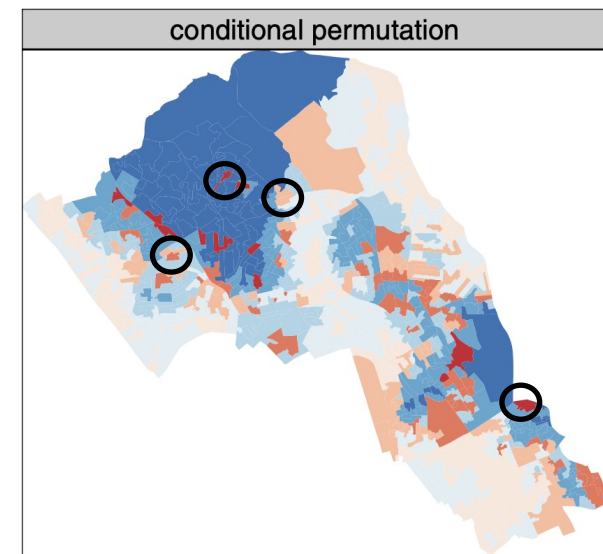
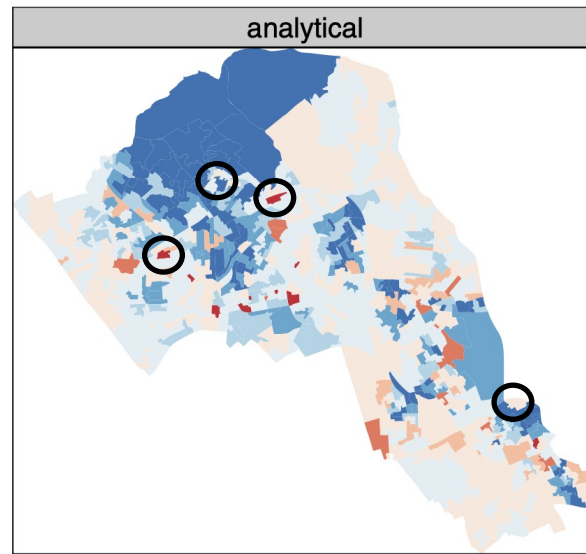
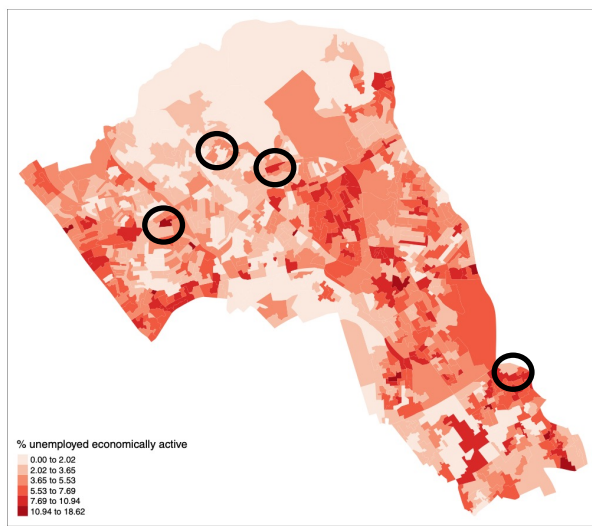
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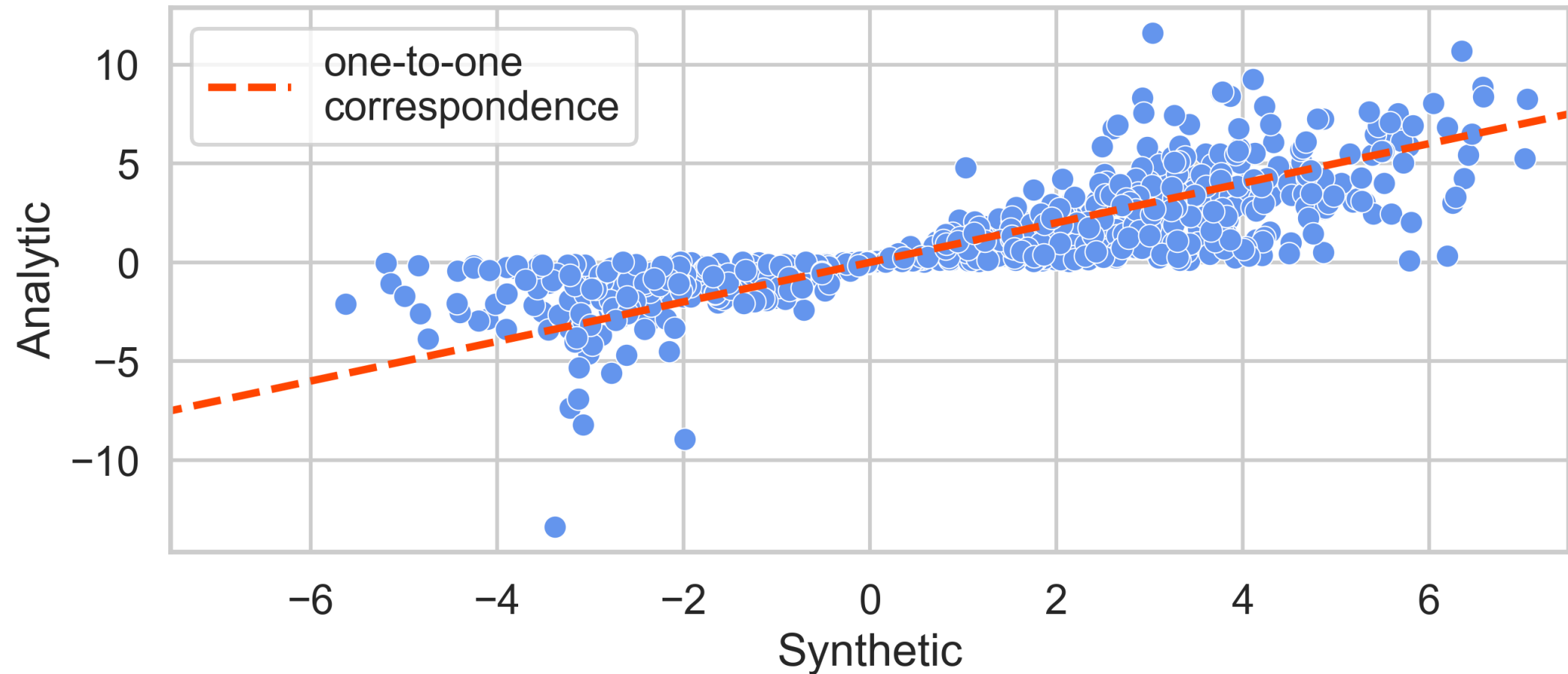
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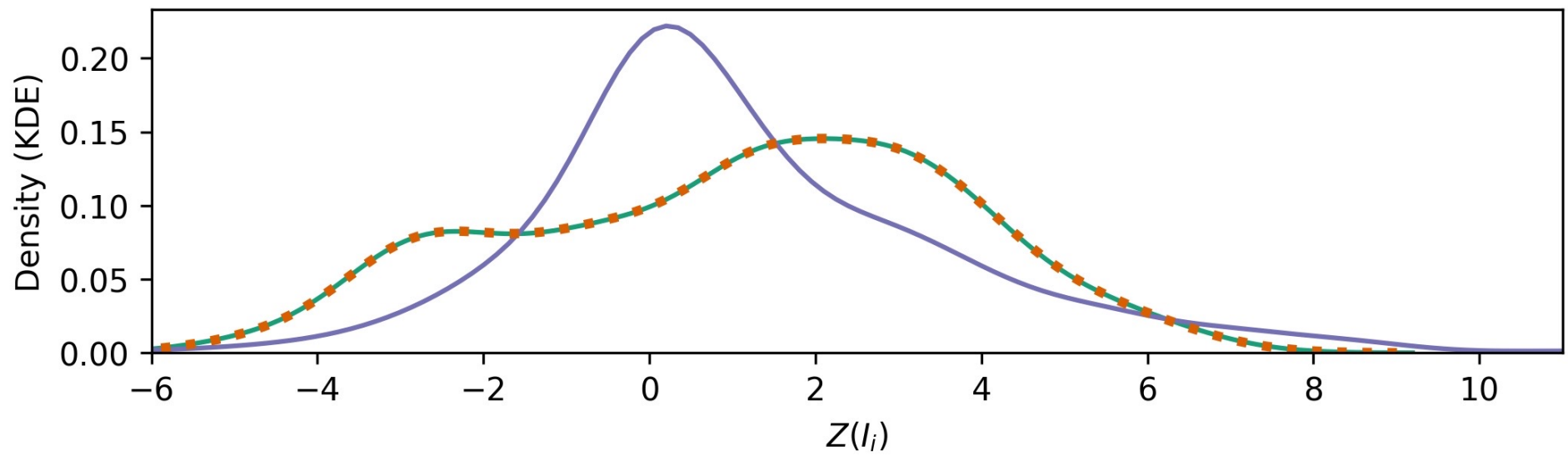
The permutation-based local z-scores were much more likely to be statistically significant [...] Analytical inference yielded 267 significant OAs whereas permutation-based inference yielded 429. [...] OAs were statistically significant using analytical inference but not using permutation-based inference, and vice versa.



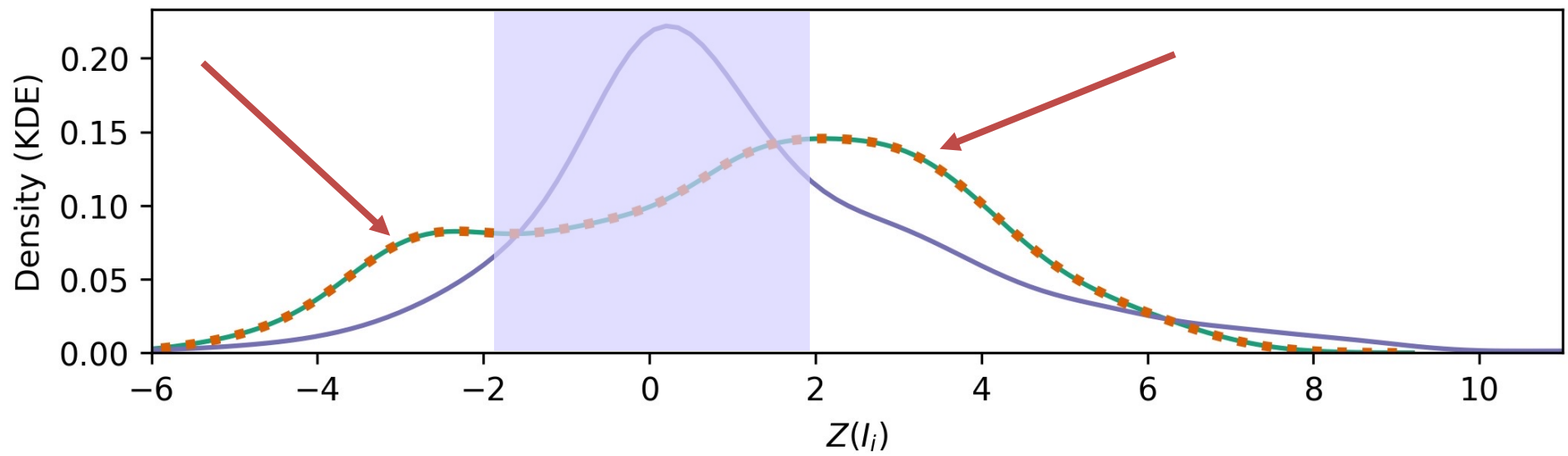
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geographical analysis

Geographical Analysis (2021) 0, 1–17

The Importance of Null Hypotheses: Understanding Differences in Local Moran's I_i under Heteroskedasticity

Jeffery Sauer¹ , Taylor Oshan¹, Sergio Rey², Levi John Wolf³

¹Center for Geospatial Information Science, Department of Geographical Sciences, University of Maryland, College Park, MD USA, ²Center for Geospatial Sciences, University of California Riverside, Riverside, CA USA, ³School of Geographical Sciences, University of Bristol, Bristol, U.K.



doi.org/10.1111/gean.12304

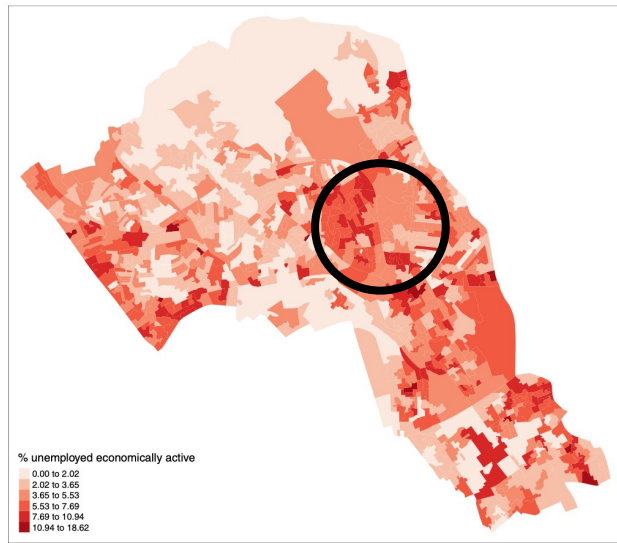
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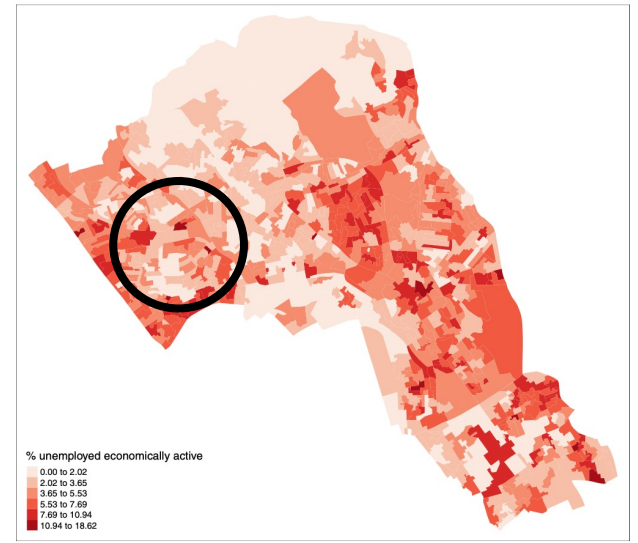


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LOW
 σ

HIGH
 σ



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“NULLS” IN LOCAL TESTS
ALL DEPENDS ON THEORY

The importance of null hypotheses in local statistics

ANALYTIC

SYNTHETIC

ANALYTIC

Assume normal observations
distributed over a regular grid

Derive $E[\Gamma]$ while assuming “all
values are equally likely at any site”

Use to define $\text{Var}[\Gamma]$

Construct z-score

Assess against a standard normal
distribution: $|z_i| > 1.96$

Hope it generalizes!

SYNTHETIC

ANALYTIC

$$\mathbf{E}[I_i] = - \sum_{j \neq i} \frac{w_{ij}}{n-1}$$

$$\begin{aligned} \mathbf{Var}[I_i] = & \frac{w_{i(2)}(n-b_2)}{n-1} \\ & + \frac{2w_{i(kh)}2b_2-n}{(n-1)(n-2)} \\ & + \left[\sum_{i \neq j} \frac{w_{ij}}{n-1} \right]^2 \end{aligned}$$

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SYNTHETIC

Assume the data that you *do have* is like data you *could have*

Hold the value at site i constant

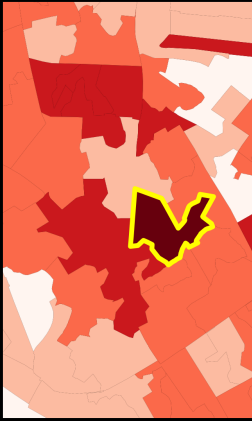
Shuffle the rest of the map

Compute a new local statistic

Repeat $k - 1$ times to obtain distribution of replicates, $\{\Gamma'\}$

Assess Γ against $\{\Gamma'\}$, so p is % of $\{\Gamma'\}$ at least as extreme as Γ

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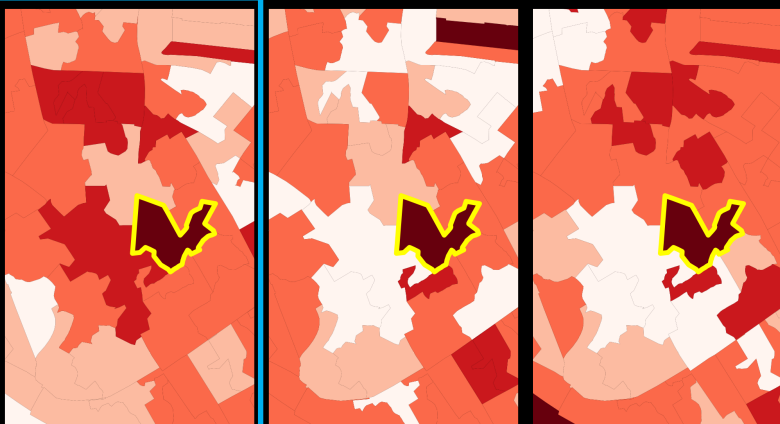
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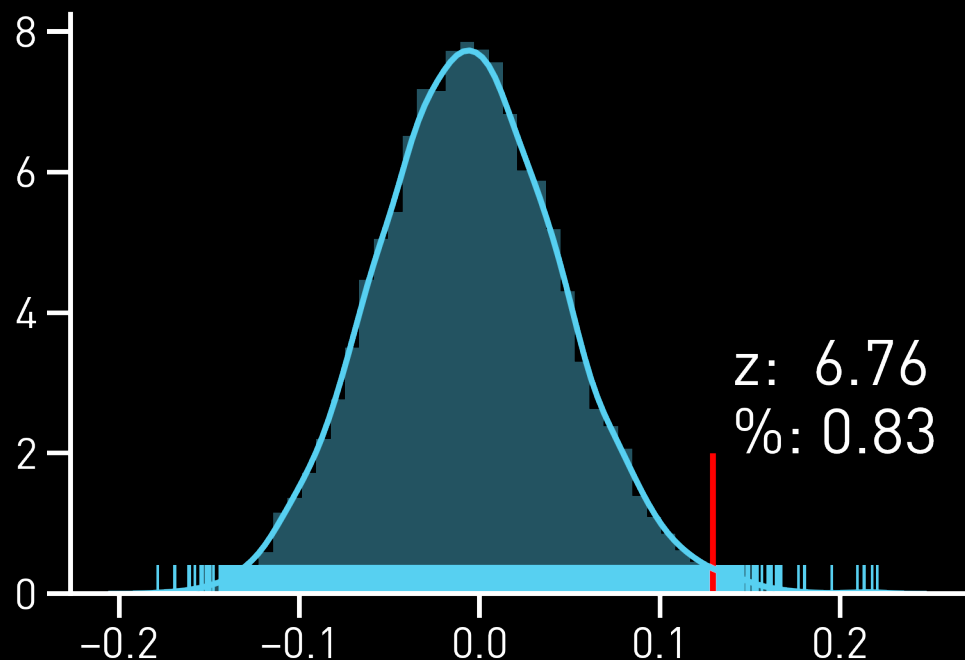
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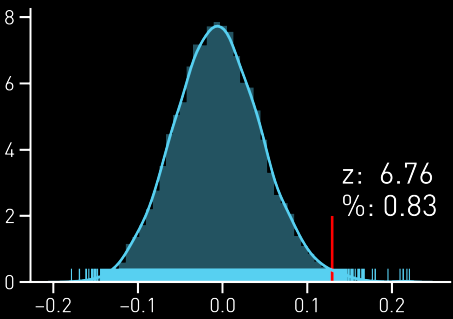
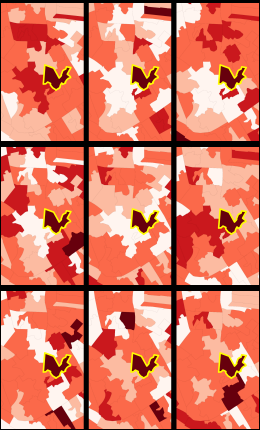
	"Total" Randomization Null <i>shuffle all values, so each value is equally likely at any site</i>	"Conditional" Randomization Null <i>for each site, the site's value is fixed and remaining sites are shuffled</i>
Analytical estimator <i>closed-form mathematical expressions for test statistics</i>	Originally in Anselin (1995). Implemented in pysal and spdep . Used in Bivand & Wong (2018). Denoted here as $E[I_i]$ and $Var[I_i]$	Originally in Sokal (1998). Not implemented in pysal and spdep or considered by Bivand & Wong (2018). Denoted here as $E_c[I_i]$ and $Var_c[I_i]$
Permutation estimator <i>test statistics computed from set of simulated maps</i>	Not considered or implemented.	Originally in Anselin (1995). Implemented in pysal . Used in Bivand and Wong (2018). Denoted here as $E_p[I_i]$ and $Var_p[I_i]$

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TOTAL

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CONDITIONAL

$$\mathbf{E}_c[I_i] = -z_i^2 \sum_{i \neq j} \frac{w_{ij}}{n-1}$$

$$\begin{aligned} \mathbf{Var}_c[I_i] = & \left[\frac{z_i}{m_2} \right]^2 \left[\frac{n}{n-2} \right] \\ & \left[w_{i(2)} - \frac{\left(\sum_{i \neq j} w_{ij} \right)^2}{n-1} \right] \\ & \left[m_2 - \frac{z_i^2}{n-1} \right] \end{aligned}$$

TOTAL

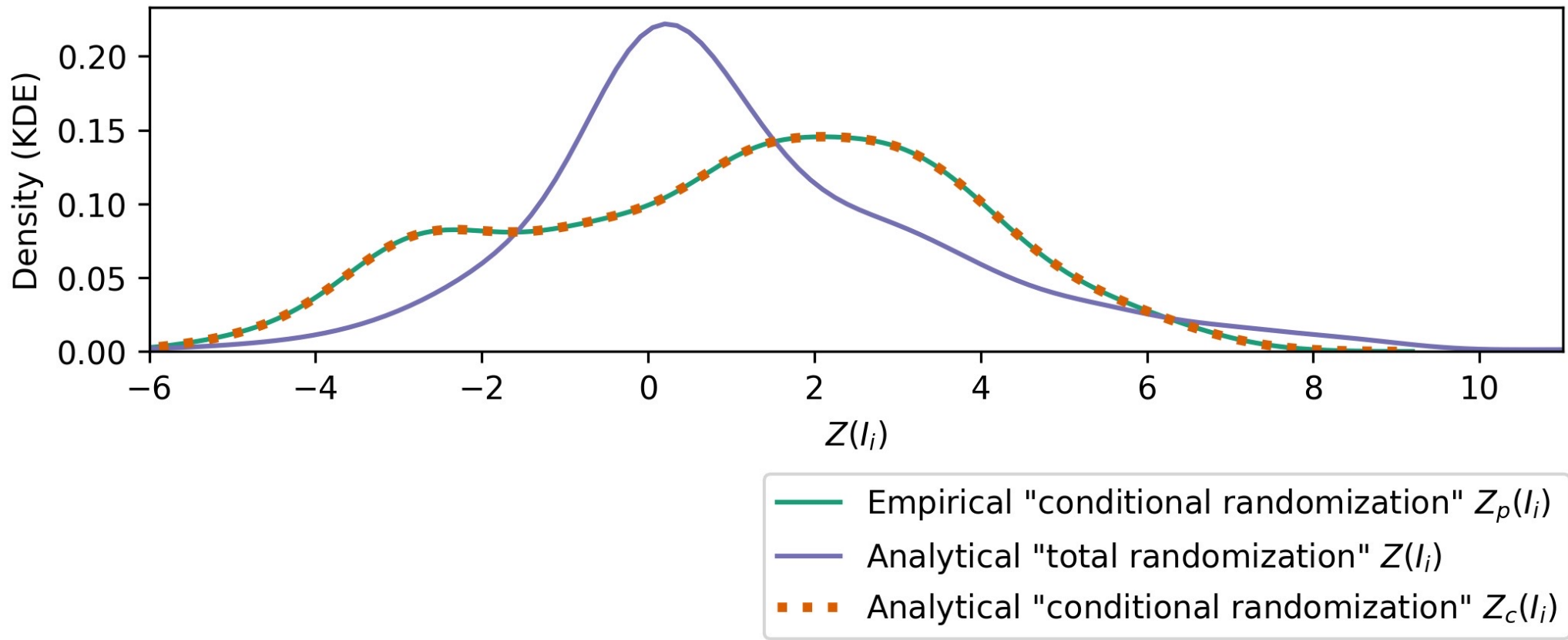
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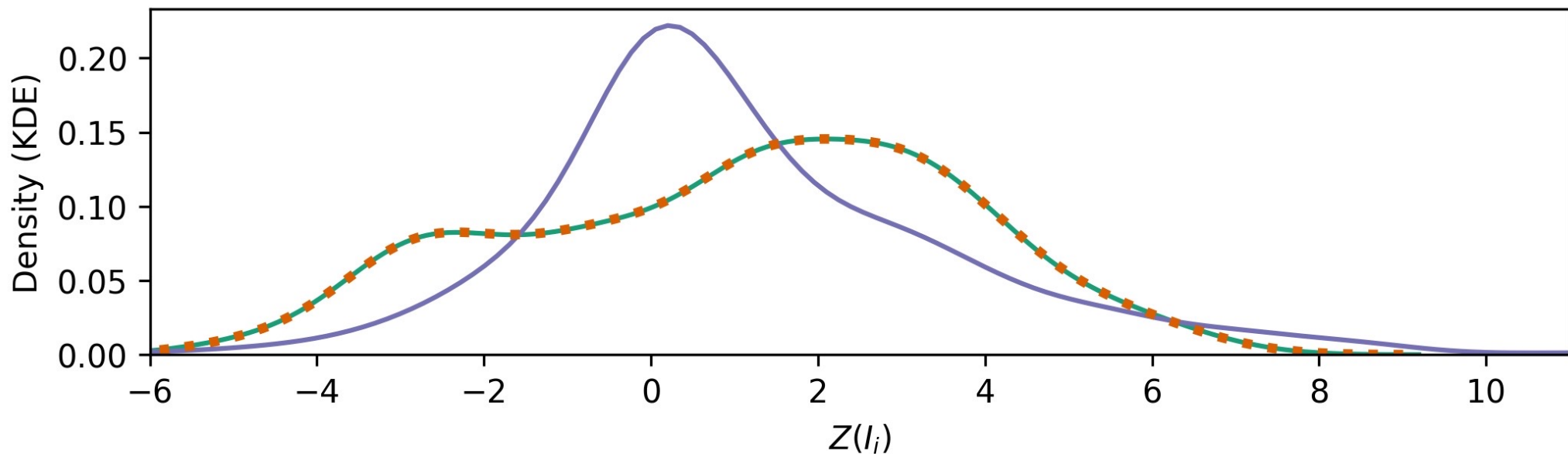
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$$\mathbf{E}_c[I_i] = - \boxed{z_i^2} \sum_{i \neq j} \frac{w_{ij}}{n-1}$$

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Perfect match for Bivand & Wong (2018)'s data



	$Z(I_i)$	$Z_p(I_i)$	$Z_c(I_i)$
$Z(I_i)$	1.00	0.85	0.85
$Z_p(I_i)$	0.85	1.00	1.00
$Z_c(I_i)$	0.85	1.00	1.00

- Empirical "conditional randomization" $Z_p(I_i)$
- Analytical "total randomization" $Z(I_i)$
- - - Analytical "conditional randomization" $Z_c(I_i)$



And practically never differ for random simulations

The Importance of Null Hypotheses: Understanding Differences in Local Moran's I_i under Heteroskedasticity



Jeffery Sauer¹ , Taylor Oshan¹, Sergio Rey², Levi John Wolf³

*A vast body of work has expanded upon Anselin (1995). But, we [still] tend to distinguish the “total” and “conditional” null by **computation**, not **conceptualization**. Our null hypotheses should be more explicitly acknowledged, however, because **they are themselves design decisions**.*

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ALL DEPENDS ON THEORY

The importance of null hypotheses in local statistics

WHERE TO?

LOCAL STATISTICS
FOR EVALUATING INTERIORITY
A REPLICATION FAILURE
FINDING "REAL" CLUSTERS
"NULLS" IN LOCAL TESTS
ALL DEPENDS ON THEORY

The importance of null hypotheses in local statistics

We do not have the right balance

- a. Expecting something to replicate &
- b. Reflexively arguing about its
“complexity” when it doesn’t

Replication across space and
time must be weak in the social
and environmental sciences

Michael F Goodchild & Wenwen Li
doi.org/10.1073/pnas.2015759118

“Democratizing” methods
makes our defaults “political!”

- Stats is a “science of defaults”
- These issues weren’t ever solved, just
supplanted by newer shinier problems

**TESTING FOR LOCAL SPATIAL
AUTOCORRELATION IN THE
PRESENCE OF GLOBAL
AUTOCORRELATION**

J. Keith Ord & Arthur Getis
doi.org/10.1111/0022-4146.00224

MUCH ADO ABOUT NULL THINGS

A replication issue in spatial stats

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